MCO-1D MARK 1 METACHRONAL CASCADE ORNITHOPTER

FLIGHT TEST CARD & PHYSICAL MEASUREMENT FIELD BINDER

Authoritative Engineering Data Collection Worksheets for Airframe Assembly, Bench Testing & Flight Trials

Worksheet Designation	Title & Scope	Hardware Milestone	Pages
WS-00	As-Built Physical Airframe & Structural Inspection	Mechanical Assembly Baseline	2 Pages
WS-01	Test Card TC-01: Ground Bench Test	Rigid Load Stand Qualification	1 Page
WS-02	Test Card TC-02: Tethered Lift & Hover	Vertical Guide-Wire Gantry	1 Page
WS-03	Test Card TC-03: Moving Vehicle Advance-Ratio Sweep	Truck Pylon 6-Axis Load Cell	1 Page
WS-04	Test Card TC-04: Free Flight STOL Runway Expansion	Airfield Taxi, STOL & Circuit Sorties	2 Pages

Demonstrator Program: MCO-1A Subscale SAR Flight Demonstrator

Hardware Baseline: 8-Wing Tandem Cascade | 10:1 Capstan Reduction | 5.687 kg MTOW

Field Protocol: Print worksheets on standard Letter paper. Complete all fields in ballpoint ink. Photograph or scan completed sheets and upload directly into the engineering console.

WS-00: As-Built Physical Airframe & Structural Worksheet

STATIC MECHANICAL BASELINE & AS-BUILT DIMENSIONS (COMPLETE BEFORE INITIAL POWER-ON)

AIRCRAFT: MCO-1D Mark 1	**CARD ID:** WS-00	**DATE:** ____ / ____ / 2026	**LOCATION:** _____
OPERATOR / TECH: _____	**AIRFRAME S/N:** MCO-1D-001	**AMBIENT TEMP:** _____ °C	**BARO / QNH:** _____ hPa

1. Total Airframe Mass Properties & Battery Inventory

Weigh all items on a calibrated digital scale (0.1 g resolution). Record actual values.

Item Description	Location / Component	Nominal Target	As-Built Measurement	Tolerance / Limit
Total Assembled Flying Mass	All 8 wings + 6S & 2S LiPos installed	*5,687.0 g (55.77 N)*	[ ] g	Max Limit: 5,950 g
Dry Airframe Mass	Both LiPo packs removed	*4,972.0 g*	[ ] g	Max: 5,100 g
6S Main Motor LiPo Pack	Tattu 6S 4500mAh 60C XT90-S	*570.0 g*	[ ] g	± 15 g
2S Receiver/Servo LiPo Pack	Gens Ace 2S 3000mAh XT30	*145.0 g*	[ ] g	± 8 g
Carbon Fuselage Spine Assembly	20mm tube + 4 station brackets	*710.0 g*	[ ] g	± 25 g
Flight Control & Avionics Core	Pi 4B + Hat + Pixhawk 4 Mini + Harness	*320.0 g*	[ ] g	± 15 g
Undercarriage & Wheels	Carbon spring struts + Du-Bro wheels	*185.0 g*	[ ] g	± 10 g

2. Individual Wing Panel Weights (S1223 Carbon / LW-PLA Finished Panels)

Bare panels with spar, ribs, covering, tip ballast slug, and root crush ferrule. Must balance port/starboard within 1.5 g.

Wing Station	Port Panel (g)	Starboard Panel (g)	Target Nominal	Port / Stbd Delta	Check (< 1.5 g)
Station 1 (W_1 Fore)	[ ] g	[ ] g	*68.5 g*	[ ] g	[ ] PASS [ ] FAIL
Station 2 (W_2 Mid-Fore)	[ ] g	[ ] g	*66.8 g*	[ ] g	[ ] PASS [ ] FAIL
Station 3 (W_3 Mid-Aft)	[ ] g	[ ] g	*65.5 g*	[ ] g	[ ] PASS [ ] FAIL
Station 4 (W_4 Aft)	[ ] g	[ ] g	*63.2 g*	[ ] g	[ ] PASS [ ] FAIL
TOTALS (All 8 Wings)	**Sum P:** [ ] g	**Sum S:** [ ] g	***528.0 g Total***	**Net: [ ] g**	**Status: [ ] OK**

WS-00: As-Built Physical Airframe (Continued — Page 2)

GEOMETRY, LONGITUDINAL STAGGER, DRIVETRAIN & BALANCE

3. Wing Semispans & Chord Measurements (Pivot Centerline to Tip)

Wing Station	Port Semi-Span (mm)	Stbd Semi-Span (mm)	Nominal Span	Root Chord (mm)	Tip Chord (mm)
Station 1 (W_1)	[ ] mm	[ ] mm	*772.0 mm*	[ ] mm (120)	[ ] mm (120)
Station 2 (W_2)	[ ] mm	[ ] mm	*749.0 mm*	[ ] mm (120)	[ ] mm (120)
Station 3 (W_3)	[ ] mm	[ ] mm	*735.0 mm*	[ ] mm (120)	[ ] mm (120)
Station 4 (W_4)	[ ] mm		*712.0 mm*	[ ] mm (120)	[ ] mm (120)
Station 1 Total Wingspan	**Tip-to-Tip Span:** [ ] mm		*1,584.0 mm Target*	Fuselage W: [ ] mm	*40.0 mm Target*

4. Cascade Stagger, Drivetrain & Dyneema Rigging Verification

Inspection Parameter	Measurement Method / Tool	Nominal Design Target	As-Built Measurement	Safety Criteria
Station 1 Pivot Axis (x_1)	Steel rule from nose tip	*138.0 mm aft*	[ ] mm	± 1.0 mm
Inter-Station Stagger (s)	Center-to-center pivot distance	*138.0 mm (or 250mm path)*	[ ] mm	± 0.5 mm
Vertical Staircase Step (z)	Vertical drop per station down spine	*41.25 mm drop*	[ ] mm	Matches CAD step
Inter-Station Convergent Slot Gap	Feeler gauge between TE flat & LE tube	*25.0 mm at neutral sweep*	[ ] mm	23.0 to 27.0 mm
10:1 Capstan Gear Ratio	Motor turns per full sector arc	*10.00 : 1 Ratio*	[ ] : 1	**Must equal 10:1**
Dyneema SK99 48h Tension	Inline hanging scale (all 8 capstans)	*180.0 N ± 5 N settled*	[ ] N	**Min 160N / Max 200N**
Spar Aluminum Crush Ferrules	Visual check: Ø8x40mm ferrule glued	*Installed all 8 spar roots*	[ ] VERIFIED ALL 8	**Mandatory (PHY-02)**
Motor Mount Thermal Gaskets	Visual: 0.5mm FR4/silicone barrier	*Fitted on all 8 motor mounts*	[ ] VERIFIED ALL 8	**Mandatory (PHY-04)**

5. Center of Gravity (CG) Envelope & Battery Keel Sled Position

Keel Sled Hole Engaged	Measured CG from Nose Tip	Calculated % MAC	Nominal Design Target	Aft Hole 6 Lockout Roll-Pin
Hole # [ ] (Target: 3)	[ ] mm	[ ] % MAC	*26.5% MAC (311.8 mm)*	**[ ] INSTALLED (Locked)**

GATE CERTIFICATION: WS-00: AS-BUILT AIRFRAME MECHANICAL CERTIFICATION	**STATUS: [ ] PASS [ ] CONDITIONAL [ ] FAIL / RE-TEST**
INSPECTOR / LEAD: _____	**SIGNATURE & DATE: _____**
REMARKS / DISCREPANCIES: _____	

WS-01: Test Card TC-01 — Ground Bench Test Worksheet

STATIC THRUST, ELECTRICAL POWER & THERMAL EQUILIBRIUM (RIGID LOAD STAND)

AIRCRAFT: MCO-1D Mark 1	**CARD ID:** TC-01	**DATE:** ____ / ____ / 2026	**LOCATION:** _____
OPERATOR / TECH: _____	**AIRFRAME S/N:** MCO-1D-001	**AMBIENT TEMP:** _____ °C	**BARO / QNH:** _____ hPa

1. Flapping Frequency Sweep & Static Performance Matrix

Mount airframe securely to rigid load stand. Execute frequency steps sequentially. Record peak readings during final 10s of dwell.

Pt	Cmd Freq	Stroke Amp	Dwell	Target Thrust	Measured Thrust	GS Bus Current	Peak Motor Temp	ODrive FET Temp	Cable Slip OK?
P1	1.0 Hz	± 60.0°	30 s	*≥ 8.0 N*	[ ] N	[ ] A	[ ] °C	[ ] °C	[ ] OK
P2	1.5 Hz	± 70.0°	30 s	*≥ 16.0 N*	[ ] N	[ ] A	[ ] °C	[ ] °C	[ ] OK
P3	2.0 Hz	± 80.0°	30 s	*≥ 24.0 N*	[ ] N	[ ] A	[ ] °C	[ ] °C	[ ] OK
P4	2.5 Hz	± 80.0°	45 s	*≥ 31.0 N*	[ ] N	[ ] A	[ ] °C	[ ] °C	[ ] OK
P5	3.0 Hz	± 80.0°	60 s	*≥ 37.0 N*	[ ] N	[ ] A	[ ] °C	[ ] °C	[ ] OK
P6	3.5 Hz	± 70.0°	20 s	*≥ 42.0 N*	[ ] N	[ ] A	[ ] °C	[ ] °C	[ ] OK

2. Electrical Subsystem, Servo Rail & Structural Dynamics Checks

Check / Parameter Description	Sensor / Measuring Point	Target Pass Criteria	Measured Value	Evaluation
7.4V Servo Bus Rail Loaded Voltage	Oscilloscope across 7.4V rail at P5	*≥ 7.10 V (droop ≤ 0.3 V)*	[ ] V	[ ] PASS [ ] FAIL
16-Servo Simultaneous Reversal Surge	Current clamp on 2S battery lead	*≤ 45.0 A peak transient*	[ ] A	[ ] PASS [ ] FAIL
6S Main Bus Voltage at Full 3.5 Hz Load	ODrive vbus telemetry	*≥ 21.60 V (no severe sag)*	[ ] V	[ ] PASS [ ] FAIL
Dyneema Max Angular Slippage	Witness mark alignment on sector rim	*≤ 0.5° (zero visible shift)*	[ ] °	[ ] PASS [ ] FAIL
Airframe RMS Accelerometer Vibration	Pixhawk 4 Mini IMU log (10–200 Hz)	*≤ 2.0 g RMS*	[ ] g	[ ] PASS [ ] FAIL
Wing Spar Deflection at Root under P6	Optical deflection scale at station yoke	*No delamination, crack, or slip*	[ ]	[ ] PASS [ ] FAIL

GATE CERTIFICATION: TC-01 GROUND BENCH TEST QUALIFICATION	**STATUS: [ ] PASS [ ] CONDITIONAL [ ] FAIL / RE-TEST**
INSPECTOR / LEAD: _____	**SIGNATURE & DATE:** _____
REMARKS / DISCREPANCIES: _____	

WS-02: Test Card TC-02 — Tethered Lift & Hover Worksheet

VERTICAL GUIDE-WIRE EQUILIBRIUM & ATTITUDE STABILIZATION AUTHORITY

AIRCRAFT: MCO-1D Mark 1	**CARD ID:** TC-02	**DATE:** ____ / ____ / 2026	**LOCATION:** _____
OPERATOR / TECH: _____	**AIRFRAME S/N:** MCO-1D-001	**AMBIENT TEMP:** _____ °C	**BARO / QNH:** _____ hPa

1. Vertical Force Equilibrium Discovery (Weightless Unstick Point)

Aircraft suspended on low-friction 4mm vertical steel guide cable. Advance frequency slowly in 0.05 Hz increments until aircraft lifts off ground stand and floats weightlessly (T/W = 1.00).

Equilibrium Parameter	Expected Range	Actual Measured Value	Engineering Notes / Telemetry Source
Weightless Unstick Frequency ($f_{ ext{hover}}$)	*1.85 to 2.10 Hz*	[ ] Hz	Frequency where load cell reaches 0.0 N tare
Commanded Flapping Amplitude	*± 75.0° (150° total stroke)*	[ ] °	Stroke angle setting on ODrive controllers
Total 6S Bus Current at Steady Hover	*18.0 to 24.0 A*	[ ] A	Summed current across all 8 ODrive inverters
Total Electrical Power at Steady Hover	*400 to 530 W*	[ ] W	$P_{ ext{elec}} = V_{ ext{bus}} imes I_{ ext{bus}}$ (Target: ~475 W)
Guide Wire Tare Collar Friction	*< 2.0 N friction*	[ ] N	Measured by lifting aircraft manually with fish scale
LiDAR Height at Steady Hover Hold	*1.0 to 1.5 m AGL*	[ ] m	TF-Luna ground altitude reading

2. Attitude Stabilization & Control Authority (Step Stick Injections)

While suspended in steady hover trim, inject ±5° stick pulses from RC transmitter / Ground Station. Verify angular response rates from Pixhawk gyro telemetry.

Control Channel	Commanded Injection	Min Required Rate	Measured Rate	Response Character / Damping
Pitch Up (AoA Fore/Aft Diff)	+ 5.0° Elevator Stick	*≥ 15.0 °/s*	[ ] °/s	[ ] Deadbeat [ ] Light Osc [ ] Sluggish
Pitch Down (AoA Fore/Aft Diff)	- 5.0° Elevator Stick	*≥ 15.0 °/s*	[ ] °/s	[ ] Deadbeat [ ] Light Osc [ ] Sluggish
Roll Right (Port/Stbd Amp Diff)	+ 5.0° Aileron Stick	*≥ 20.0 °/s*	[ ] °/s	[ ] Deadbeat [ ] Light Osc [ ] Sluggish
Roll Left (Port/Stbd Amp Diff)	- 5.0° Aileron Stick	*≥ 20.0 °/s*	[ ] °/s	[ ] Deadbeat [ ] Light Osc [ ] Sluggish
Yaw Right (Fore/Aft Sweep Diff)	+ 10.0° Rudder Stick	*≥ 12.0 °/s*	[ ] °/s	[ ] Deadbeat [ ] Light Osc [ ] Sluggish
Yaw Left (Fore/Aft Sweep Diff)	- 10.0° Rudder Stick	*≥ 12.0 °/s*	[ ] °/s	[ ] Deadbeat [ ] Light Osc [ ] Sluggish

GATE CERTIFICATION: TC-02 TETHERED LIFT & HOVER QUALIFICATION	**STATUS: [ ] PASS [ ] CONDITIONAL [ ] FAIL / RE-TEST**
INSPECTOR / LEAD: _____	**SIGNATURE & DATE:** _____
REMARKS / DISCREPANCIES: _____	

WS-03: Test Card TC-03 — Moving Vehicle Advance-Ratio Sweep

AERODYNAMIC POLAR MAPPING, FORWARD THRUST & LIFT (TRUCK TEST PYLON)

AIRCRAFT: MCO-1D Mark 1	**CARD ID:** TC-03	**DATE:** ____ / ____ / 2026	**LOCATION:** _____
OPERATOR / TECH: _____	**AIRFRAME S/N:** MCO-1D-001	**AMBIENT TEMP:** _____ °C	**BARO / QNH:** _____ hPa

1. Aerodynamic Force & Polar Sweep vs Forward Airspeed

Mount aircraft to 6-axis load cell pylon above open-bed truck. Drive runway in steady speed steps in calm air (< 1.0 m/s wind).

Run	Truck Speed (V_{inf})	Flap Freq	Thrust ($3F_{xS}$) Target	Measured $3F_{xS}$	Target Lift ($3L_S$)	Measured Lift	Pitch Moment ($3M_{yS}$)	Pitot Speed
R1	4.0 m/s (9 mph)	2.20 Hz	*≥ 12.0 N*	[ ] N	*≥ 20.0 N*	[ ] N	[ ] N·m	[ ] m/s
R2	6.2 m/s (14 mph)	2.30 Hz	*≥ 15.0 N*	[ ] N	*≥ 56.0 N (Unstick)*	[ ] N	[ ] N·m	[ ] m/s
R3	9.0 m/s (20 mph)	2.40 Hz	*≥ 8.0 N*	[ ] N	*≥ 56.0 N (Cruise)*	[ ] N	[ ] N·m	[ ] m/s
R4	12.0 m/s (27 mph)	2.60 Hz	*≥ 5.0 N*	[ ] N	*≥ 58.0 N*	[ ] N	[ ] N·m	[ ] m/s
R5	14.0 m/s (31 mph)	2.80 Hz	*≥ 2.0 N (Sprint)*	[ ] N	*≥ 62.0 N*	[ ] N	[ ] N·m	[ ] m/s

2. Pure Passive Glide Aerodynamics Verification (Wings Stationary: f = 0.0 Hz)

Hold wings completely stationary at clean cruise pitch (α = +4.0°). Measures baseline fixed-wing lift deficit vs airspeed.

Truck Speed (V_{inf})	Flapping State	Expected Passive Lift	Measured Passive Lift	Measured Drag ($3D_S$)	Measured $3L/3D_S$ Ratio	Flight Safety Note
9.0 m/s (20.1 mph)	**0.0 Hz (Fixed)**	*37.13 N (18.6 N deficit)*	[ ] N	[ ] N	[ ]	Cannot sustain 1g level flight
11.0 m/s (24.6 mph)	**0.0 Hz (Fixed)**	*55.77 N (1.00g balance)*	[ ] N	[ ] N	[ ]	**Safe unpowered dead-stick glide**
13.0 m/s (29.1 mph)	**0.0 Hz (Fixed)**	*77.80 N (excess lift)*	[ ] N	[ ] N	[ ]	Positive climb glide flare

3. Pitot Differential Pressure Sensor vs True Truck Ground Speed

Truck GPS Speed: 6.2 m/s	**Target Counts:** ~56 counts (23.5 Pa)	**MS4525DO Counts:** [ ]	**Indicated Airspeed:** [ ] m/s
Truck GPS Speed: 9.0 m/s	**Target Counts:** ~118 counts (49.6 Pa)	**MS4525DO Counts:** [ ]	**Indicated Airspeed:** [ ] m/s

GATE CERTIFICATION: TC-03 ADVANCE-RATIO TRUCK SWEEP QUALIFICATION	**STATUS: [ ] PASS [ ] CONDITIONAL [ ] FAIL / RE-TEST**
INSPECTOR / LEAD: _____	**SIGNATURE & DATE:** _____
REMARKS / DISCREPANCIES: _____	

WS-04: Test Card TC-04 — Free Flight STOL Sortie Worksheet

FIRST FREE-FLIGHT SORTIE: TAXI, UNSTICK, CLIMB, CRUISE, FLARE & ROLLOUT

AIRCRAFT: MCO-1D Mark 1	**CARD ID:** TC-04	**DATE:** ____ / ____ / 2026	**LOCATION:** _____
OPERATOR / TECH: _____	**AIRFRAME S/N:** MCO-1D-001	**AMBIENT TEMP:** _____ °C	**BARO / QNH:** _____ hPa

1. Pre-Flight Electrical, Sensor & Environmental Go/No-Go Gate

System / Parameter	Inspection Requirement	Go/No-Go Safety Limit	Pre-Flight Measurement	Status
6S Main Motor LiPo Voltage	DMM resting check at battery lead	*≥ 25.0 V (≥ 4.17 V/cell)*	[ ] V	[ ] GO [ ] NO-GO
2S Receiver/Servo LiPo Voltage	DMM check at servo bus tap	*≥ 8.30 V (≥ 4.15 V/cell)*	[ ] V	[ ] GO [ ] NO-GO
Surface Wind Speed	Handheld anemometer on runway	*≤ 3.0 m/s aligned with runway*	[ ] m/s	[ ] GO [ ] NO-GO
Crosswind Component	Crosswind perpendicular to centerline	*≤ 1.0 m/s max crosswind*	[ ] m/s	[ ] GO [ ] NO-GO
Battery Sled Position Lock	Verify quick-pin in Hole 3 (250 mm)	*Hole 3 locked, Hole 6 pin in place*	[ ] VERIFIED HOLE 3	[ ] GO [ ] NO-GO
Pitot Cover Removed & Zeroed	Pixhawk telemetry indicated airspeed	*0.0 m/s ± 0.2 m/s calm*	[ ] m/s	[ ] GO [ ] NO-GO

2. Takeoff Acceleration & Climb Performance (Phase 1)

Flight Parameter	Telemetry Channel / Source	Design Predicted Target	Actual Flight Recorded	Tolerance / Target
Takeoff Ground Roll Distance	Runway visual markers / GPS	*≤ 12.0 m ground roll*	[ ] m	STOL target: 9–11 m
Unstick Indicated Airspeed ($V_{ ext{rot}}$)	Pitot airspeed at wheels-off	*6.20 m/s (13.9 mph)*	[ ] m/s	± 0.4 m/s
Elapsed Time to Unstick	Time from advance lever to liftoff	*3.5 to 4.5 seconds*	[ ] s	< 5.0 s limit
Steady Rate of Climb (V_z)	Pixhawk barometer / EKF3 climb rate	*1.80 to 2.20 m/s at 9 m/s*	[ ] m/s	Boyd P_s envelope
Flight Path Climb Angle (gamma)	gamma = arcsin(V_z / V_inf)	*11.5° to 15.0° climb*	[ ] °	Clearance obstacle
Peak Electrical Takeoff Power	6S bus current × voltage log	*≤ 680 W total peak*	[ ] W	< 32 A total

WS-04: Free Flight STOL Sortie (Continued — Page 2)

CRUISE HANDLING, APPROACH, FLARE, TOUCHDOWN & POST-FLIGHT DEBRIEF

3. Level Cruise Flight & Handling Qualities (Phase 2)

Cruise Metric	Telemetry Source / Parameter	Nominal Design Target	Actual Flight Recorded	Performance Window
Level Cruise Airspeed (V_inf)	Pitot calibrated airspeed (CAS)	*9.00 m/s (20.1 mph)*	[ ] m/s	8.5 to 9.5 m/s
Active Cruise Flapping Frequency	ODrive commanded frequency	*2.20 to 2.50 Hz*	[ ] Hz	**Floor: ≥ 1.87 Hz**
Cruise Throttle Lever Position	RC transmitter throttle channel	*48% to 55% detent*	[ ] %	Cruise detent
Mean Electrical Power Consumption	Average bus power in steady level	*240 to 280 W*	[ ] W	< 300 W limit
Specific Energy (Cost of Transport)	Wh energy consumed per km	*7.4 to 8.6 Wh/km*	[ ] Wh/km	< 9.5 Wh/km
Pitch Trim Stick Bias	Elevator trim offset for level flight	*0.0% ± 5.0%*	[ ] %	Validates 26.5% CG
Roll Trim Stick Bias	Aileron trim offset for straight track	*0.0% ± 3.0%*	[ ] %	Validates wing balance

4. Approach, Flare, Touchdown & Rollout (Phase 3)

Landing Metric	Target Expected	Actual Flight Recorded	Safety Evaluation & Notes
Final Approach Indicated Airspeed	*7.50 m/s ± 0.3 m/s*	[ ] m/s	Stabilized glideslope approach
Flare Airbrake Command Response	*Clamshell high-drag flare*	[ ] CLEAN FLARE [ ] SLOW	H-54 / CONV-BRK deceleration
Touchdown Airspeed (Wheels-On)	*4.50 to 5.20 m/s*	[ ] m/s	Gentle wheel contact
Landing Ground Rollout Distance	*≤ 8.5 m from touchdown to stop*	[ ] m	Du-Bro wheel rollout braking
Peak Vertical Touchdown Load	*≤ 1.8 g vertical spike*	[ ] g	Gear damping performance

5. Post-Flight Battery Recovery & Thermal Health Inspection

Flight Duration: [ ] min : [ ] sec	**6S Rested Voltage: [ ] V (≥ 21.8V)**	**Capacity Consumed: [ ] mAh**
2S Rested Voltage: [ ] V (≥ 7.4V)	**Max Motor Temp: [ ] °C (≤ 65°C)**	**Max ODrive FET Temp: [ ] °C (≤ 75°C)**
Dyneema Lines Post-Flight: [ ] ZERO FRAYING [ ] WITNESS MARKS INTACT [ ] TENSION VERIFIED (180N)		

GATE CERTIFICATION: TC-04 FREE FLIGHT ENVELOPE EXPANSION CERTIFICATION	**STATUS: [ ] PASS [ ] CONDITIONAL [ ] FAIL / RE-TEST**
INSPECTOR / LEAD: _____	**SIGNATURE & DATE: _____**
REMARKS / DISCREPANCIES: _____	